

Generation Interconnection System Impact Study Report

For Queue Project Q15-0524

Sierra Blanca 100 kV

October 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0524. The Interconnection Customer proposes a 143.4 MW solar photovoltaic facility interconnecting at the Sierra Blanca 100 kV substation in the Illinois area. The request was received on October 19, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 285 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,224,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CONRAD AUTO to SIERRA BLANCA SUBSTATION 100 kV Line	WALLENPAUPACK to MINNEAPOLIS CITY OF 500 kV	135.2	125	100.0
CLINTMOORE to CONRAD AUTO 100 kV	MINNEAPOLIS CITY OF to WALLENPAUPACK 500 kV line	133.4	123	100.0
BEAR CANYON to CLINTMOORE 100 kV	CLINTMOORE to LAKE CHARLES PLANT 100 kV	128.7	121	100.0
CLINTMOORE to LAKE CHARLES PLANT 100 kV	BEAR CANYON to CLINTMOORE 100 kV line	119.5	130	100.0
ARCH SMITH to SWEATT 100 kV Circuit 2	ARCH SMITH to SWEATT 100 kV Ckt 1	110.5	189	9.1
HOOSIER FAIRVIEW to SPEARFISH HYDRO 100 kV	SPEARFIS HYDRO to HOOSIER FAIRVIEW SUBSTATION 100 kV	108.5	146	5.2
SPEARFISH HYDRO to HOOSIER FAIRVIEW 100 kV #2	HOOSIER FAIRVIEW to SPEARFISH HYDRO 100 kV Line	108.5	146	5.2
BEAR CANYON SUBSTATION to SOCRATES 19 100 kV line	CLINTMOORE to LAKE CHARLES PLANT 100 kV	106.7	129	100.0
SWEATT to ARCH SMITH 100 kV Cir. 1	SWEATT to ARCH SMITH 100 kV Cir. 2	103.9	201	9.1
MYRTLE PT to GALION GENERTING STA 230 kV Line	TREGO to BEAVER 500 kV	103.6	828	7.5
LAWRENCE HILL to SWEATT 100 kV	BLYTHE to SOUTH BOSTON 230 kV line	102.3	194	8.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CONRAD AUTO to SIERRA BLANCA 100 kV Line	Rebuild line with higher-capacity conductor	\$4,483,000	100% (\$4,483,000)
CLINTMOORE to CONRAD AUTO 100 kV	Rebuild line with higher-capacity conductor	\$2,973,000	100% (\$2,973,000)
BEAR CANYON to CLINTMOORE 100 kV	Rebuild line with higher-capacity conductor	\$4,255,000	100% (\$4,255,000)
CLINTMOORE to LAKE CHARLES PLANT 100 kV	Rebuild line with higher-capacity conductor	\$6,342,000	100% (\$6,342,000)
ARCH SMITH to SWEATT 100 kV Cir. 2	Rebuild line with higher-capacity conductor	\$1,604,000	17% (\$278,000)
HOOSIER FAIRVIEW to SPEARFISH HYDRO 100 kV	Reconductor line	\$472,000	100% (\$472,000)
HOOSIER FAIRVIEW to SPEARFISH HYDRO 100 kV line	Reconductor line	\$939,000	100% (\$939,000)
SOCRATES 19 to BEAR CANYON SUBSTATION 100 kV L	Reconductor line	\$1,133,000	100% (\$1,133,000)
ARCH SMITH to SWEATT 100 kV Cir. 1	Reconductor line	\$699,000	17% (\$121,000)
MYRTLE PT to GALION GENERATING STA 230 kV Line	Reconductor line	\$10,419,000	1% (\$124,000)
LAWRENCE HILL to SWEATT 100 kV	Reconductor line	\$833,000	12% (\$104,000)
	Total allocated to project		\$21,224,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.